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A systematic review identifying fitness metrics indicative of simulated astronaut extravehicular activity task performance

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Nicole C. Strock [1,2] ✉, Alyssa N. Varanoske [1,2], Jakob R. Allen [1,2] & Dillon J. Frisco [2,3]

To prepare for safe and successful completion of surface operations during future missions to the Moon and Mars, understanding the relationships between fitness and simulated extravehicular activity (EVA) performance is imperative. This systematic review included twelve studies [366 astronaut and non-astronaut subjects (206 male, 154 female, 6 unspecified)] that reported cardiorespiratory or muscular fitness data in relation to EVA task performance and/or simulated EVA intensity outcomes. Databases searched (through May 2025) included PubMed, Web of Science, Scopus, and NASA Technical Reports Server. The AXIS critical appraisal tool for cross-sectional studies assessed study quality. Results were synthesized in tabular and graphical formats. Thirty-eight fitness metrics were identified, with 8 cardiovascular and 12 muscular metrics demonstrating associations with simulated EVA task performance outcomes. This systematic review identified fitness metrics of relevance for future exploration EVAs and provides important considerations for exercise countermeasure and monitoring capability development. Funded by NASA Human Research Program.

For the first time in over 50 years, NASA's Artemis program is preparing for a sustained human presence on the Lunar surface (NASA's Plan for Sustained Lunar Exploration and Development), bringing new opportunities and challenges for spaceflight exploration. One such challenge will be the need to physically prepare astronauts for regular Lunar Extravehicular Activities (EVAs; up to 5 EVAs for short-duration missions, or 3–4 per week for sustained missions)[1,2], with projected Artemis Notional EVA timelines planning for sustained effort of up to 8 h (6 ± 2 h)[2]. Frequency and duration of Artemis EVAs will exceed previous Apollo experiences, which averaged approximately 4 h per EVA and 3 EVAs per mission (NASA EVA Data). In combination with a larger exploration spacesuit[3,4], the demands associated with Artemis partial-gravity (g) EVAs can be expected to exceed those of previous Apollo EVAs and microgravity EVAs performed on the International Space Station (ISS)[5], as tasks are expected to be of greater complexity.

Exposure to microgravity manifests in well-documented systemic physiological deconditioning, with potential to yield performance-limiting impacts[6]. Therefore, the maintenance of physical fitness (i.e., cardiorespiratory and muscular) through exercise is necessary for crewmembers to remain healthy enough to carry out nominal and off-nominal operations. Understanding which elements of fitness may be most indicative of EVA performance will provide critical information for physical preparation prior to and during surface exploration missions to help ensure safe and successful completion of Lunar EVA mission objectives. Planned exploration EVAs will involve science, engineering, and maintenance objectives that may require a range of physical workloads to ambulate, utilize tools and collection equipment, deploy science packages, and transport/stow materials[1]. Additionally, critical tasks and contingency scenarios, compounded with the expected performance deficits due to spaceflight deconditioning, may require greater physical fitness. Studies employing simulated EVA tasks performed at high efforts provide an indication of workloads and fitness necessary for the successful completion of tasks in challenging and potentially off-nominal situations.

Prioritizing health and fitness represents an attainable risk mitigation strategy for planetary EVAs[7]. To ensure that astronauts are physically prepared for Artemis missions and that appropriate in-mission exercise countermeasures and monitoring capabilities are provided[8], exploration fitness standards should be developed and validated based on the expected objectives and tasks. Currently, fitness standards for partial-g extraterrestrial surface EVAs introduced in the most recent NASA 3001 Standards (NASA-STD-3001) include thresholds for aerobic capacity (VO_{2max}; 36.5 mL/kg/min) and muscular strength (1-Repetition Maximum [1-RM] bench press: 1.0 × body weight; 1-RM deadlift: 1.6 × body weight); however, these have not been validated with respect to Artemis-associated EVA operations. Additionally, exploration vehicle mass and volume constraints may limit the equipment available to monitor fitness. Therefore, investigating alternative fitness metrics may help guide future monitoring capability needs for

[1]KBR, Inc, Houston, TX, USA. [2]Exercise Physiology and Countermeasures Laboratory, NASA Johnson Space Center, Houston, TX, USA. [3]JES Technologies, Houston, TX, USA. ✉e-mail: nicole.c.strock@nasa.gov

NASA, other spaceflight agencies, and private companies developing standards for exploration vehicles and exercise countermeasures.

While Apollo missions represented an impressive feat of testing and expansion of human space exploration capabilities yielding critical lessons learned[9] that influenced the development of exploration programs, many questions remain regarding the physical factors associated with safe and successful EVA performance. Until humans return to the Moon and set foot on Mars, research involving 1 g and partial-g offload analog simulations provide invaluable evidence to inform the optimization of human performance and EVA readiness. As such, the purpose of this systematic review of literature is to explore the relationship between fitness characteristics and simulated EVA task performance. A consolidation of findings from previous 1 g and analog investigations can be used as a basis for future research to validate exploration fitness standards in relation to EVA task performance for Artemis design reference mission objectives.

Methods

The present systematic review of literature followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines[10,11] and was conducted to address the following research question: "How does fitness (cardiorespiratory and muscular) relate to functional task performance for EVAs or EVA simulations?"

Information sources and search strategy

Potential studies for inclusion were first identified using PubMed, NASA Technical Reports Server (NTRS), and a hand search. The literature search was performed in November–December 2023, following the search strategy in Table 1. A second round of searching was performed in April–May 2025, to update the previous search and include Web of Science and Scopus. There were no restrictions on publication dates, but language was limited to English. Due to the lack of Boolean logic for the NTRS, keywords from search number 2 were utilized.

Eligibility criteria

For inclusion in this review, published manuscripts and technical reports needed to meet criteria for population, intervention, control conditions, outcomes and study design (PICOS) criteria, ultimately including fitness data in relation to EVA task performance and/or simulated EVA intensity outcomes (Table 2). Exclusion criteria included review papers and manuscripts that did not report any fitness outcomes in relation to EVA task performance and/or simulated EVA intensity outcomes. No consent was required as the study is a systematic review.

Selection process

Studies were screened and reviewed by two reviewers (NS, AV) using the Rayyan web app for systematic reviews[12]. Citations first underwent an initial title and abstract screening to exclude studies that did not address the study research question or meet PICOS criteria. Full-text articles were retrieved for studies that passed initial screening. For discrepancies, a third reviewer was consulted (JA).

Data collection process and data items

Data was extracted and managed via an Excel spreadsheet. Data included subject demographics (number of participants, sex, age, body mass), description of methods employed by authors for task circuit, standalone task, EVA simulation (descriptions, suited or gravity condition, pace, intensity, failure), fitness metrics, identified thresholds for task performance (e.g., fitness thresholds associated with task completion/failure or time to completion), and reported associations/relationship to performance outcomes. Data were summarized in tabular and graphic formats.

Study risk of bias assessment

Risk of bias and quality of study design and reporting were assessed using the AXIS critical appraisal tool for cross-sectional studies[13]. This tool was chosen as it more closely applied to the design of studies included in this review. The 20-component tool was used to determine the quality and risk of bias of a study based on study design and reporting considerations. For each component question, reviewers answered "yes", "no", or "do not know" based on the information included within the study. Two reviewers (N.S., A.V.) completed the assessment independently, and a third reviewer (J.A.) completed the assessment for specific components where there was disagreement in scoring to serve as a tie-break. A figure was compiled based on the consensus of responses; because several components were not directly applicable to studies, as they pertained to "non-responders" and reviewers unanimously scored these components as "do not know" for each study, these components were not included in the final figure.

Results

The process for study selection and exclusion is outlined in Fig. 1. The search for studies identified 786 citations, of which 12 were removed due to duplication, yielding a total of 774 citations for screening. After title and abstract screening, 748 citations did not meet the eligibility criteria and were excluded. A full-text review of the remaining 26 citations excluded 14 additional studies due to a lack of fitness assessment presented related to EVA task performance[14–22] or EVA task circuit/simulation[23–26], or wrong publication type[27]. A total of 12 studies[28–39] (Table 3) were included in this review, which reported fitness outcomes in relation to EVA task performance and/or simulated EVA intensity outcomes. In total, these studies included 366 astronaut and non-astronaut subjects (206 M, 154 F, 6 unspecified), with sample sizes ranging from 6 to 71 subjects and average age of subjects ranging from 23 to 47 years (where reported).

Of these studies, 6 utilized designated task circuits/testing[28–30,33,34,36], 7 investigated standalone EVA tasks[28,31,32,35,37–39], and 2 incorporated an EVA simulation[32,37] (Table 3). A majority of studies (11/12) were conducted in 1 g conditions, 1 study included Lunar and Martian offload conditions[39], and 2 studies were performed at 1 g with Lunar (20 mL/kg/min[30]; targeting

Table 1 | Pubmed search strategy

Search number	Term	Key words	Search mask
1	Extravehicular activity simulation	"Extravehicular activity" [Mesh] OR "EVA" OR "space walk" OR "space-walk"	Title/Abstract
2	Task and physical performance	"Task performance" OR "physical performance" OR "mission task" OR "Functional task" OR "performance"	Title/Abstract
3	Environment	"Hypogravity" [Mesh] OR "low gravity" OR "reduced gravity" OR "altered gravity" OR "partial gravity" OR "lunar gravity" OR "moon gravity" OR "Lunar" OR "moon" OR "martian gravity" OR "mars gravity" OR "mars" OR "Weightlessness" [Mesh] OR "zero gravity" OR "microgravity"	All fields
4	Fitness	"Cardiorespiratory fitness" [Mesh] OR "muscle strength" OR "endurance"	All fields
5	Exercise testing	"Exercise test" [Mesh] OR "fitness testing" OR "cardiopulmonary exercise test" OR "physical fitness testing" OR "treadmill test" OR " bicycle ergometry test" OR "exercise tolerance" [Mesh]	All fields
6	Combined search	1 AND 2 AND (3 OR 4 OR 5)	All fields

Table 2 | PICOS inclusion criteria

	Description
Population	Terrestrial subjects (astronauts as applicable), no gender or age restrictions
Interventions	Terrestrial simulated extravehicular activities or EVA-operationally relevant tasks, including unsuited/shirt sleeve 1 g, microgravity 0 g, suited (1 g, simulated Lunar 1/6 g, Martian 3/8 g)
Control conditions	Terrestrial (1 g) where applicable
Outcomes	*Aerobic fitness*: Peak aerobic capacity, ventilatory threshold, critical speed, critical power, peak power output, anaerobic power
	Muscular fitness: isokinetic strength peak torque and total work (concentric/eccentric), maximal isometric force (leg press, bench press, mid-thigh pull; knee extension), isotonic power (leg press, bench press), countermovement/vertical jump
	EVA intensity: aerobic capacity, heart rate, rate of perceived exertion, energy expenditure
Study designs	All types of experimental designs were included

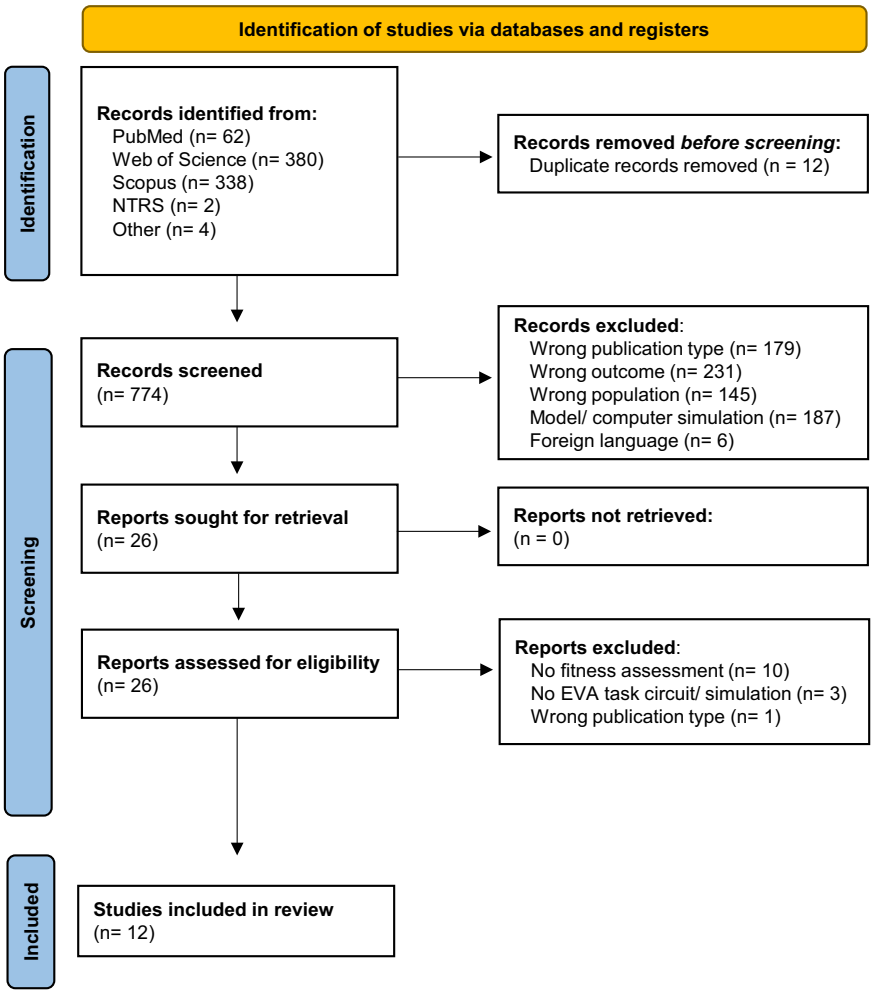


Fig. 1 | PRISMA flow diagram. NTRS NASA Technical Reports Server, EVA extravehicular activity.

17–20 mL/kg/min[35]) and Martian (30 mL/kg/min[30]; pacing with the addition of weights added to torso and extremities[35]) intensities achieved through specified ranges of aerobic expenditure.

For EVA task circuits, subjects performed a variety of EVA-inspired tasks, including those requiring large muscle groups such as object carry/transport, climbing (treadmill hill, stairs, ladders), seat egress, hatch opening, and emergency crew rescue, as well as fine motor tasks such as manipulating connections and tool use (drilling, wrenching) (Table 3). The number of distinct tasks included in a task circuit ranged from 3 to 6 distinct tasks, performed in a 1 g shirt-sleeve condition, with the exception of 3 studies which utilized externally weighted suits[33,34] or a weighted pack with distributed weights[36]. In 5 of 6 task circuit studies, subjects completed circuits as fast as possible, with average completion time ranging from 20 to 36 min. One task circuit employed pacing strategies to mimic Lunar or Martian EVA VO_2 intensities[30]. Two task circuit studies required subjects to complete multiple rounds of the circuit, ranging from 5 to 15 rounds[29,30]. Three task circuit studies reported task circuit failure rates[29,30,34].

Of the studies included, 7 of 12 included standalone tasks, such as a traverse (ranging from 1.5 to 10 km)[28,35,39], hill climb[32,35,37], or capsule egress[31]. Three studies reported time to completions for standalone traverse tasks (10 km: 61–95 min[28,39]) or capsule egress (54.9 ± 19.4 s[31]). One study looking at standalone tasks used pacing strategies to mimic Lunar or Martian EVA VO_2 intensities[35].

Of the EVA simulation and task circuit studies reviewed, only 6 involved subjects performing EVA tasks in some form of a suit or suit simulator[32–34,36,37,39]. Of the suited studies, 1 included simulated 1/6 g (Lunar)

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Table 3 | Study demographics

Study	*N* (M/F)	Age (yr)	Body mass (kg)	Category	Description	Suited condition	Gravity condition	Pace	Actual intensity	Task fail
Ade et al.[29]	70 (39 M/ 31 F)	23 ± 5	Not reported	Task circuit	**Material transport field test (all participants): 10x** (1) Material loading- loaded 10 × 10 kg simulated core samples into a 2-wheeled cart designed to simulate the modular equipment transporter used by the Apollo 14 astronauts (2) Material transport- pushed cart across a 54 m level track (3) Material unloading- removed the simulated core samples and returned them to a table **Device operations field test (24 participants): 5x** (1) Valve and bolt manipulation- turned 4 valves + bolts (5 cm) against a resistance which was provided by springs. (2) Hose couplings- connected and disconnected 4 × 7.6 cm fire hose couplings (3) Balance beam walk - traversed a 10 cm wide x 3.66 m long balance beam (4) Equipment drag-pulled a weighted sled 18 m with a rope	Shirtsleeve (athletic apparel)	1 g	As fast as possible	Not reported	**Material transport field test**: 19/70 (27%) **Device operations field test**: Not reported
Ade et al.[30]	21 (10 M/ 11 F)	26 ± 7	76.0 ± 15.7	Task circuit	**Field test (10 m between each task):** (1) Ladder climb (2) Agility cones (3) Stair climb (4) Lateral climb (5) Equipment lift - moved 2 × 4.5 kg boxes from waist to eye level and 2 × 9 kg boxes from floor to waist level and back (6) Step-entry lateral movement with 40 cm hurdles and ducked under chest-level poles	Shirtsleeve (athletic apparel)	1 g	**Lunar**: 20 mL/kg/min **Martian**: 30 mL/kg/min	**Lunar**: ~19 mL/kg/min (~44% VO_{2pk}) **Martian**: ~28 mL/kg/min (~65% VO_{2pk})	**Lunar**: All completed **Martian**: 6/21 (28%); could only complete 65% of the Martian test
Ryder et al.[34]	60 (32 M/ 28 F)	37 ± 7 (24–52)	75 ± 13.3	Task circuit	(1) Capsule egress: supine to stand, ladder deploy, lift and push the un-deployed life raft (13.6 kg) and 2 crew survival packs (6.4 kg each), climb out of the hatch (2) Ambulation and supply transfer (1.5 km walk + 30 (10.9 kg) crew transfer bag carries 5 m) (3) Rescue drag (mannikin (75 kg) rolled onto rescue sled, dragged 50 m) (4) Hill climb and descent (2020 m (2–8% grade))	Five external load conditions in the weighted suit	1 g	As fast as possible	Not reported	**Capsule egress**: 2/ 290 trials **Ambulation/ transfer**: 7/290 trials **Rescue drag**: 2/ 290 trial **Hill climb**: 28/283 trials

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Table 3 (continued) | Study demographics

Study	*N* (M/F)	Age (yr)	Body mass (kg)	Category	Description	Suited condition	Gravity condition	Pace	Actual intensity	Task fail
Ryder et al.[33]	20 (10 M/10 F) 6 (Shuttle Crew; 4 M/ 2 F)	36 ± 10 (25–54)	68.8 ± 14	Task circuit	(1) Seated egress and walk (once from upright, once from supine + obstacle course) (2) Recovery from fall to stand (prone to standing) (3) Hatch opening test (peak force turning wheel for 3–5 s; then turn wheel at 50% max torque for 20 s) (4) Passive ladder climb test (40 rungs) (5) Object translation (pick up and move tools, equipment, samples- 2.7–9 kg over 2.4 m)	Three external load conditions in the weighted suit per group	1 g	As fast as possible	Not reported	Not reported
Taylor et al.[36]	8 (5 M/3 F)	34.9 ± 3.7	72.9 ± 8.3	Task circuit	(1) Hatch opening: Biodex upper extremity wheel attachment (2) Hand drilling: Biodex lateral rotating pinch attachment (3) Construction wrenching: Biodex upper extremity wrench attachment (4) Inclined walk: 0.8 km on a treadmill (angle of inclination 1° every 2 min until it reached 5 degrees, decline at the same rate down to 0°) (5) Sample collection: collect 9 medicine balls arranged in two successive rows at 3 and 6 m from a designated starting position; retrieve and deposit in a storage rack (6) Emergency crewmember drag: drag a 54 kg dummy across 13.5 m (8.5 m straight, 90 degree turn, 5 m drag)	Equipped with weighted backpack plus distributed weights (48 kg) to simulate the NDX-2 space suit (ankle = 0.07 kg, chest 11.34 kg, waist = 9.07 kg, wrist = 2.29 kg, backpack = 15.88 kg)	1 g	As fast as possible	**Hatch opening**: VO_2: 24.8 ± 4.4; RPE: 6 9 ± 2.3 **Hand drilling**: VO_2: 18.5 ± 3.0; RPE: 5.5 ± 2.1 **Construction wrenching**: VO_2: 28.3 ± 6.3; RPE: 6.8 ± 1.9 **Inclined walk**: VO_2: 32.3 ± 6.6; RPE: 6.3 ± 2.0 **Sample collection**: VO_2: not reported; RPE: 4.3 ± 1.5 **Emergency drag**: VO_2: not reported ; RPE: 4.8 ± 1.7	Not reported
Ade et al.[28]	71 (40 M/31 F)	23 ± 5	73.8 ± 15.7	Task circuit	**Physical abilities field test (15 rounds):** (1) Ladder climb (12 ft ladder, 10 rungs) (2) Agility cones, forward and backward movement through 6 vertical poles 6 ft apart (3) Stair climb, 4.5 m set of stairs, each step 25 cm rise (4) Lateral climb (5) Box lift, 2 × 4.5 kg boxes from waist to eye level, then 2 × 9 kg boxes from floor to waist (6) Step over task, moved laterally and stepped over 40 cm high hurdles and ducked under chest-level poles	Shirtsleeve (athletic apparel)	1 g	As fast as possible	Not reported	Not reported
				Standalone task	10 km Walkback					

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Table 3 (continued) | Study demographics

Study	*N* (M/F)	Age (yr)	Body mass (kg)	Category	Description	Suited condition	Gravity condition	Pace	Actual intensity	Task fail
Alexander et al.[31]	15 (5 M/ 10 F)	47 ± 4	90 ± 23	Standalone task	**Capsule egress test:** seated supine, roll over, move 2 × 5 kg bags 1 m, release and attach rope ladder to floor, carry 2 packages through top hatch, exit capsule through top hatch	Shirtsleeve (athletic apparel)	1 g	As fast as possible	VO_2 reached 72 ± 25% of relative VO_{2pk}	No failures
Norcross et al.[39]	6 (6 M)	46.8 ± 4.3 (40–51)	81.4 ± 7.8	Standalone tasks	Treadmill ambulation (POGO)	POGO: shirtsleeve with PLSS backpack mock-up [18 kg] + gimbal support system [41 kg]	1/6 g	Self-paced walking and running	**Unsuited**: 8.8–16.6 mL/kg/min **Suited:** 16.7-29.3 mL/kg/min	Not reported
							3/8 g		**Unsuited**: 11–12.1 mL/kg/min **Suited**: 27.6-35.7 mL/kg/min	
					10 km walkback (0% grade treadmill)	Mark III suit (MKIII)	1/6 g	Self-paced walking and running	**Avg %VO_{2pk}**: 50.8 ± 6.1% (45–61%) **Avg speed:** 1.74 ± 0.22 m/s (6.3 km/h) **Avg VO_2**: 2.0 ± 0.3 L/min **Total energy expenditure**: 944.2 ± 70.5 kcal **RPE**: 11.8 ± 1.6	All completed within 2 h
Sutterfield et al.[35]	45 (20 M/ 25 F)	39 ± 10	79 ± 20	Standalone tasks	(1) 1500 m treadmill traverse (constant speed 0.85 m/s; on track with synthetic rocks and pebbles to simulate Mars surface) followed by 30 × 10 lb bag carries for 10 m each (2) 2000 m up/down Hill climb (0–8% grade; 55–99 m/min or 3.3–5.9 km/h) carrying 5.5 kg body bar on ascent, no bar on descent	Shirtsleeve (athletic apparel)	1 g	**Low intensity**: 17–20 mL/kg/min	**Hill climb**: 20 ± 2.3 mL/kg/min (~56% VO_{2pk}) **Surface Traverse:** 24.0 ± 1.8 mL/kg/min (~67% of VO_{2pk})	**Hill climb**: 6/45 (13%) Surface Traverse: 6/45 (13%)
								Moderate intensity: Pacing + 13.6 kg (torso) + 2.3 kg (each ankle + upper arm)	**Hill Climb**: 26.0 ± 0.7 mL/kg/min (~ 73% VO_{2pk}) **Surface traverse**: 28.7 ± 1.8 mL/kg/min (~ 80% VO_{2pk})	**Hill climb**: 10/39 (25%) **Surface traverse**: 15/39 (38.5%)
Mulavara et al.[38]	13 (ISS crew; 11 M/ 2 F) 10 (BR no ex; 10 M) 9 (BR ex; 9 M)	47 ± 5 38 ± 7 34 ± 6	84 ± 14 80 ± 9 77 ± 7	Standalone tasks	(1) Seated egress and walk (2) Recovery from fall to stand (3) Object translation (pick up and move tools, equipment, samples- 2.7, 4.5, 9 kg 2.4 m) (4) Jump down test (30 cm) (5) passive ladder climb test (40 rungs) (6) Activity board: moving 2 hose connectors from their initial attachments to an adjacent attachment, moving 3 electrical-connectors from vertical positions to horizontal positions, and removing and installing a handle using a cordless power driver to loosen 2 screw (7) Hatch opening test (peak force turning wheel for 3–5 s)	Shirtsleeve (athletic apparel)	1 g	Self-paced	Not reported	Not reported

Table 3 (continued) | Study demographics

Study	*N* (M/F)	Age (yr)	Body mass (kg)	Category	Description	Suited condition	Gravity condition	Pace	Actual intensity	Task fail
Dyson et al.[37]	6 (5 M/1 F)	Not reported	Not reported	EVA simulation	"EVA" simulations: (1–2 h) designed to mimic geological surveying (hill climbing, flat terrain walking, rock hammer, quiet standing)	Mars analog suit	1 g	EVA simulation: self-paced	**Average VO_2:** 35.9 ± 0.9 mL/kg/min **HR**: 81% of max	Not reported
				Standalone Tasks	Hill runs: subjects ran up and down 200-m hill 10% until exhaustion (intended as a proxy for traditional treadmill testing)			**Hill run**: pace increased every 2 min until exhaustion	**Average VO_2:** 52.4 ± 1.0 mL/kg/min	
Rai et al.[32]	6 (M/F not reported)	23 ± 2.4	Not reported	EVA simulation	"EVA" simulations: (1–2 h) designed to mimic geological surveying (hill climbing, flat terrain walking, rock hammer, quiet standing)	Mars analog suit	1 g	EVA simulation: self-paced	EVA reached ~85% VO_{2peak} and HR_{max}	Not reported

Data reported as mean ± standard deviation.
M male, *F* female, VO_2 aerobic capacity, VO_{2pk}/VO_{2max} peak or maximal aerobic capacity, *RPE* rate of perceived exertion, *BR* bed rest, *ex* exercise.

and 3/8 g (Martian) offload[39], and 2 included data for simulated 1 g EVAs[32,37]. No studies where the goal was to determine fitness thresholds required for EVAs included a high-fidelity suited condition.

Cardiorespiratory fitness in relation to EVA tasks/simulated EVA

EVAs require sustained efforts that range in intensity depending on tasks included in surface operations timelines. For exploration missions, tasks may include constructing and maintaining outposts, deploying scientific packages, and sample collection, which may require traverse over long distances in a weighted spacesuit. Understanding the minimum level of cardiorespiratory fitness necessary to aid in safe and successful completion of mission objectives is imperative. As such, multiple studies have been conducted to evaluate which components of cardiorespiratory fitness best relate to EVA-specific task performance[28–31,34,35] (Table 4). A total of 13 cardiorespiratory fitness assessments were identified, with 8 cardiorespiratory metrics associated with EVA performance outcomes.

Correlates of task performance (Fig. 2A)

Aerobic capacity: One of the key physiological factors that influences endurance exercise performance, aerobic capacity (VO_{2pk} or VO_{2max})[40], was the most common aerobic fitness metric examined in studies with EVA task circuits or standalone tasks. This was assessed using various methods throughout the studies, including treadmill, arm crank, cycle ergometer, and rowing ergometer. Values were reported as either VO_{2pk} or VO_{2max} within each study, and consistency is maintained herein. All but 1 study[39] reported VO_{2pk} or VO_{2max} as relative to body mass (i.e., mL/kg/min). When relating VO_{2max} to completion time for various EVA task circuits[28,29], there were significant negative associations for arm crank ($r = -0.41$ to -0.66) and treadmill ($r = -0.43$ to -0.74) VO_{2max}, such that individuals with higher VO_{2max} values (i.e., greater cardiorespiratory fitness) resulted in shorter time to completion (i.e., better performance). In one study that parsed out the relationship between VO_{2pk} and completion time in male and female subjects, negative associations tended to be stronger for males than females (treadmill VO_{2max}: $r = -0.78$ [M] vs. $r = -0.60$ [F]; arm crank VO_{2max}: $r = -0.69$ [M] vs. $r = -0.52$ [F])[28].

For studies that included standalone tasks, such as a 10 km walkback, treadmill VO_{2max} ($r = -0.64$) was negatively associated with completion time[28]. Negative associations tended to be stronger for male ($r = -0.73$) vs. female ($r = -0.51$) subjects for treadmill VO_{2max} and 10 km walkback time[28]. For capsule egress, neither cycle ($p = 0.08$) nor rowing VO_{2pk} ($p = 0.07$) was correlated with egress time. Even subjects with low VO_{2pk} (≤20 mL/kg/min) were able to complete capsule egress in 2 minutes[31].

Gas exchange threshold (GET)/ventilatory threshold (VT): The GET, denoted as the non-linear increase in VCO_2 vs. VO_2, is a noninvasive method to estimate the lactate threshold[41]. Due to strong correlation with endurance performance and indication of submaximal workload[42], GET has been explored as an alternative metric for EVA-related performance. When relating GET to completion time for various EVA task circuits, there were significant negative associations for arm crank (range $r = -0.34$ to -0.51) and treadmill GET (range $r = -0.45$ to -0.58), expressed as mL/kg/min[28,29]. For certain EVA task circuits, sex differences in the relationship between GET and completion time emerged. For instance, in the physical abilities field test presented by Ade and colleagues[28], arm crank GET was significant for both male ($r = -0.48$) and female ($r = -0.47$) subjects, while treadmill GET was only significantly related to completion time in male subjects ($r = -0.73$). Additionally, while arm crank GET expressed as a percentage of VO_{2pk}, was associated with completion time for the physical abilities test ($r = -0.33$), treadmill GET was not. For standalone tasks, such as a 10 km walkback, treadmill GET ($r = -0.48$) expressed as mL/kg/min, was negatively associated with completion time[28]; however, negative association was only present in male subjects ($r = -0.59$)[28]. Treadmill GET expressed as a percentage of VO_{2pk} was not associated with completion time for 10 km walkback.

Aerobic Peak Power Output (PPO): Few studies have examined the relationship between PPO and task performance. In one study, PPO

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Table 4 | Aerobic fitness metrics

Fitness category	Fitness sub-category	Fitness metric	Thresholds	Associations/ relationship to performance outcomes
Aerobic fitness	Maximal aerobic capacity (VO_{2peak} or VO_{2max})	Cycle VO_{2peak} (mL/kg/min)	Egress: <20 to 27.5 (Alexander et al.[31]; Ryder et al.[34])	Not associated with egress (Alexander et al.[31])
			Ambulation/transfer: 27.4 (Ryder et al.[31])	
			Crew rescue: 27.41 (Ryder et al.[31])	
			Hill climb (low and moderate intensity): 26 and 30.5 (100% sensitivity; Sutterfield et al.[35]);	
			Surface traverse (low and moderate intensity): 26 and 40 (100% sensitivity; Sutterfield et al.[35])	
		Rowing VO_{2peak} (mL/kg/min)	Not reported	Not associated with egress (Alexander et al.[31])
		Treadmill VO_{2max} (mL/kg/min)	Task circuit (Martian intensity; 50% probability of completion): 38.4 (Ade et al.[30])	EVA task circuit: $r = -0.43$ to -0.74 (Ade et al.[28,29])
			Task circuit (Martian intensity; 75% probability of completion): 40.0 (Ade et al.[30])	10 km walkback: $r = -0.64$ (Ade et al.[28])
		Arm crank VO_{2max} (mL/kg/min)	Not reported	EVA task circuit: $r = -0.42$ to -0.66 (Ade et al.[28,29])
	Gas exchange threshold (GET)	Treadmill GET	Not reported	EVA task circuit: $r = -0.45$ to -0.58 (Ade et al.28,29)
				10 km walkback: $r = -0.48$ (Ade et al.[28])
		Arm crank GET	Not reported	EVA task circuit: $r = -0.34$ to -0.51 (Ade et al.[28,29])
		Cycle GET (mL/kg/min)	Surface traverse (low and moderate intensity):16 and 28.8 (100% sensitivity; Sutterfield et al.[35])	Not reported
			Hill Climb (low and moderate intensity): 14.8 and 18.7 (100% sensitivity; Sutterfield et al.[35])	
	Aerobic peak power (PPO)	Cycle PPO (W)	Surface traverse (low and moderate intensity):188 and 238 (100% sensitivity; Sutterfield et al.[35])	Not associated with egress (Alexander et al.[31])
			Hill Climb (low and moderate intensity): 188 (100% sensitivity; Sutterfield et al.[35])	
		Rowing PPO (W)	Surface traverse (low and moderate intensity):188 and 262 (100% sensitivity; Sutterfield et al.[35])	Associated with egress ($r = -0.60$) (Alexander et al.[31])
			Hill Climb (low and moderate intensity): 188 (100% sensitivity; Sutterfield et al.[35])	
		Arm crank PPO	Not reported	EVA task circuit: $r = -0.35$ to -0.56 (Ade et al.[28,29])
	Critical power (CP)	Rowing CP (W)	Surface traverse (low and moderate intensity): 99.7 and 170 (100% sensitivity; Sutterfield et al.[35])	Not associated with egress (Alexander et al.[31])
			Hill climb (low and moderate intensity):124 and 123 (100% sensitivity; Sutterfield et al.[35])	
		Arm Crank CP (W)	CP > 70.5 low risk of time in the lowest performance quartile (Ade et al.[29])	EVA task circuit: $r = -0.61$ to -0.70 (Ade et al.[28,29])
			CP ≤ 39.8 high risk of time in the lowest performance quartile (Ade et al.[29])	
	Critical speed (CS)	Treadmill CS	Not reported	EVA task circuit: $r = -0.56$ to -0.82 (Ade et al.[28,29])
				10 km walkback: $r = -0.85$ (Ade et al.[28])

Thresholds were identified as specific values determined through regression or receiver operator curves as being related to task performance.

Relationships were reported as r, r^2, or Somers' D values. VO_{2peak} peak aerobic capacity, VO_{2max} maximal aerobic capacity, *EVA* extravehicular activity, *GET* gas exchange threshold, *PPO* peak power output, *CP* critical power, *CS* critical speed.

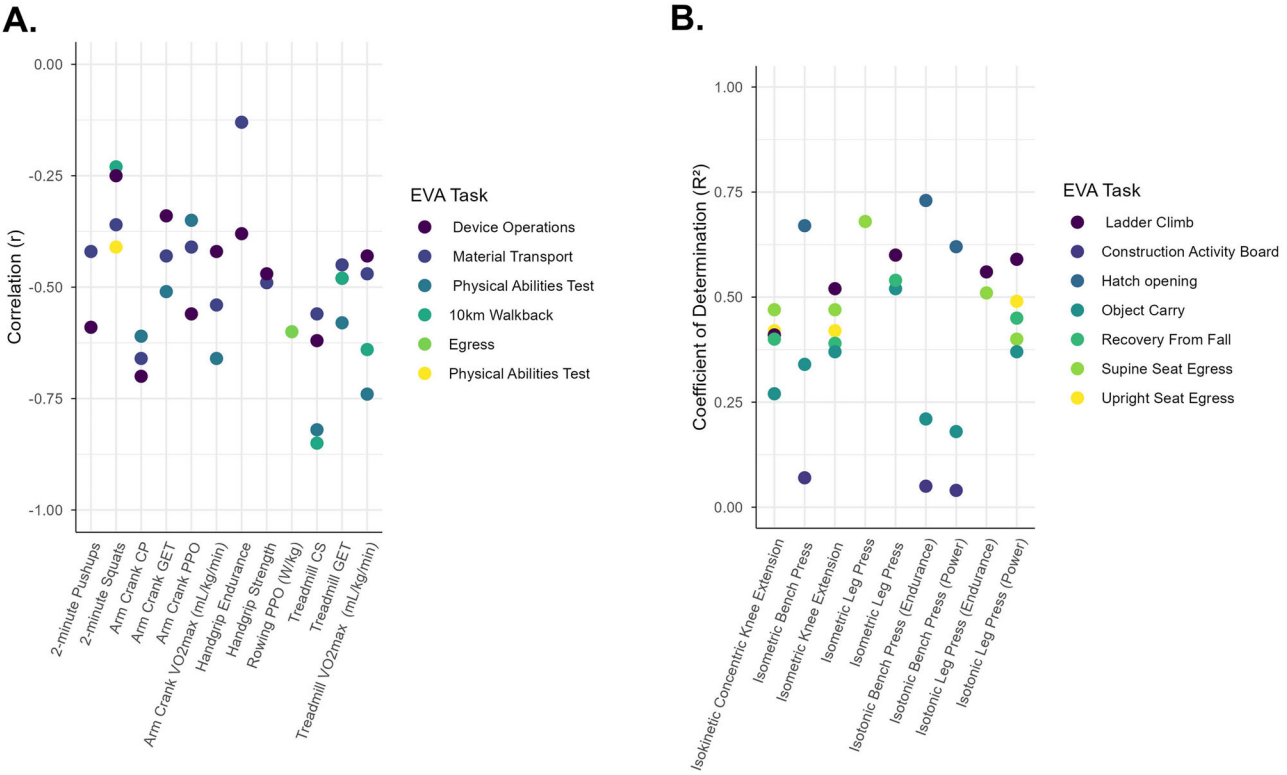


Fig. 2 | Relationships between fitness and task performance. Cardiorespiratory and muscular fitness metric **A** correlations [r] and **B** coefficient of determination [R^2] values reported in relation to Extravehicular (EVA) task circuit or standalone tasks time to completion. Correlations with more negative values indicate that the greater the fitness metric, the faster the EVA task or task circuit is completed (i.e., better performance). R^2 values closer to 1 indicate that the fitness metric accounted for more variance in the time to completion regression models for EVA tasks. CP critical power, GET gas exchange threshold, PPO peak power output, VO_{2max} maximal aerobic capacity, CS critical speed.

obtained during incremental VO_{2pk} tests assessed via cycle ergometer was not correlated with egress time ($p = 0.44$)[31]. However, egress time was inversely associated with rowing PPO normalized to kg body weight ($r = -0.60$, $p = 0.03$), indicating that the higher the PPO, the quicker the egress time[31].

Critical power/critical speed: Tests of critical power (CP) provide an indication of exercise tolerance and may inform fatigue development[43]. While CP is commonly assessed on a cycle ergometer, the power-time relationship is conserved with other modalities (e.g., running) but is presented as critical speed (CS), when intensity is assessed in units of speed instead of power[43,44]. Previous research demonstrated that CS, in comparison to GET or VO_{2max}, was a better determinant of endurance exercise performance[45,46]. Few studies examined CP or CS in relation to EVA task performance. Arm crank CP was associated with time to completion for EVA task circuits ($r = -0.61$ to -0.70[28,29]) and was found to be an independent predictor of the lowest performance quartile, yielding a predictive capability of 82.9%[29]. Treadmill CS was associated with time to completion for EVA task circuits ($r = -0.56$ to -0.82[28,29]) and 10 km walkback ($r = -0.85$[28]), and relationships were comparable among male ($r = -0.82$ to -0.88) and female ($r = -0.78$ to -0.82) subjects[28]. One study examined rowing CP and found that this metric was not associated with egress time ($p = 0.15$[31]).

Thresholds for success/completion time

Aerobic capacity*:* Several studies reported distinct thresholds, as identified through regressions, to be associated with EVA task circuit performance. In one such study, cycle VO_{2pk} thresholds for four tasks (capsule egress, ambulation and supply transfer, rescue drag, and hill climb and descent) were comparable, ranging from 27.4 to 27.6 mL/kg/min[34]. Similar results in cycle VO_{2pk} thresholds were demonstrated in an investigation by Sutterfield et al.[35], where Lunar intensities resulted in a threshold of 26 mL/kg/min, yielding 100% sensitivity for determining failure. However, Martian simulated efforts for these tasks required greater fitness as the surface traverse threshold for subjects who completed the task was 40 mL/kg/min, and hill climb was 30.5 mL/kg/min[35]. Additionally, another investigation found treadmill VO_{2pk} to be the best predictor of Martian field test success[30], although no other aerobic capacity methods were utilized. VO_{2max} values with 50% and 75% probability of finishing were 38.4 mL/kg/min and 40.0 mL/kg/min, respectively[30]. Additionally, those who maintained efforts <65% of VO_{2max} could complete the Martian field test, with non-finishers working at 78% of VO_{2max}[30].

GET/VT: When performing field tests at greater intensities (e.g., Martian), GET for non-finishers was lower than those who finished (21.9 vs. 27.1 mL/kg/min), and average field test VO_{2pk} values exceeded their GET ($p = 0.005$)[30]. Treadmill GET was an independent predictor of Martian field test failure. GET values with 50% and 75% probability of finishing were 20.1 mL/kg/min and 25.1 mL/kg/min[30]. Univariate logistic regression indicated that cycle VT was a significant predictor of task performance for surface traverse and hill climb tasks[35]. Additionally, cycle VT with 100% sensitivity for test failure was determined for surface traverse (Lunar: 16 mL/kg/min; Martian: 28.8 mL/kg/min) and hill climb tasks (Lunar: 14.8 mL/kg/min; Martian: 18.7 mL/kg/min)[35].

Aerobic PPO: Univariate logistic regression indicated that both cycle and rowing PPO were significant predictors of task performance for surface traverse and hill climb tasks[35]. Additionally, cycle and rowing PPO thresholds were determined for surface traverse (Lunar: 188 W; Martian: 238–262.5 W) and hill climb tasks (Lunar: 188 W; Martian: 188 W)[35]. Cycle thresholds had strong sensitivity for identifying EVA test failure (100% sensitivity), while also performing well to minimize the identification of false positives (e.g., correctly identifying test completion; 62.5–89.7% specificity).

Rowing thresholds had strong sensitivity for identifying EVA test failure (100% sensitivity), but lower specificity (33.3–68.9%).

Critical power/critical speed: Treadmill CS was found to be the single best predictor of 10 km walkback time, while CS and arm crank CP were predictors of field test performance time[28] and rowing CP was a significant predictor of surface traverse and hill climb tasks[35]. Rowing CP thresholds, which indicated 100% sensitivity for test failure, were determined for successful completion of surface traverse (Lunar: 99.7 W; Martian: 170 W) and hill climb (Lunar: 124 W; Martian: 123 W)[35]. Thresholds emerged for arm crank CP, such that individuals with CP > 70.5 W were at low risk for having time to completion in the lowest performance quartile, while CP < 39.8 W were at high risk of having time in the lowest performance quartile[29]. Logistic regression models including arm crank CP had 90.2% sensitivity (correctly classified 46/51) and 63.2% specificity (correctly classified non finishers 12/19)[29]. No thresholds were investigated for arm crank CP or treadmill CS.

Muscular fitness in relation to EVA tasks/simulated EVA

EVAs are considered full-body tasks, requiring astronauts to lift, carry, push, and pull equipment across varying terrains in a weighted spacesuit. As such, muscular strength, power, and endurance have been evaluated in multiple studies comparing laboratory-based strength tests to functional ability on EVA-specific task performance[28,29,33,34,36,38] (Table 5). A total of 25 muscular fitness assessments were identified (Table 5), with 12 muscular metrics associated with EVA performance outcomes.

Correlates of task performance (Fig. 2A, B)

Strength-isometric: Isometric strength is defined as the maximum voluntary force produced while in a static position. Isometric testing can be completed across a variety of muscle groups and joint positions, and is reported to be valid, reliable, and safe, making it an attractive method of assessing strength in multiple populations[47]. Maximal isometric force for leg press and knee extension accounted for at least 37% of the variance in completion time models for functional tasks primarily requiring lower body effort and ambulation, including supine and upright seat egress ($r^2 = 0.42–0.68$), ladder climb (LP: $r^2 = 0.52–0.60$), recovery from fall ($r^2 = 0.39–0.54$), and object carry ($r^2 = 0.37–0.52$; classified as both lower and upper body task)[33]. Maximal isometric force for bench press accounted for more variance in completion time models for upper body functional tasks, including hatch opening ($r^2 = 0.67$) and object carry ($r^2 = 0.34$), but less for construction activity board ($r^2 = 0.07$)[33]. One study examined correlations between handgrip strength and EVA task performance, reporting inverse relationships between handgrip maximal voluntary contraction and time to complete material transport ($r = -0.49$, $p < 0.05$) and device operations tasks ($r = -0.47$, $p < 0.01$)[29], such that a stronger grip strength resulted in quicker time to completion (i.e., better performance).

Similarly, changes in maximal voluntary isometric leg press force and functional performance outcomes in bed rest subjects and astronauts who completed 6-month-long missions aboard the ISS[38] were seen. Reductions in leg press strength were negatively associated with impairments (i.e., increased completion time) on 3 of the 7 functional tasks, including seat egress and walk ($D = -0.305$, $p < 0.001$), recovery from fall ($D = -0.363$, $p < 0.001$), and object translocation ($D = -0.285$, $p < 0.001$). Additionally, changes in leg press strength were marginally associated (after controlling for False Detection Rate and multiple associations, $0.003 < p < 0.004$) with reduced ladder climb time ($D = -0.189$, $p = 0.004$) and hatch opening strength ($D = 0.175$, $p < 0.004$). Changes in upper body strength were not significantly related to changes in any functional test outcome in this investigation ($p > 0.05$).

Strength-isokinetic: Another widely used type of strength assessment is isokinetic testing, defined as exercise that dynamically loads contracting muscle to its maximum capability while the speed of the movement remains constant throughout the entire range of motion. Maximal knee extension torque (60°/s) accounted for the most variance in models for completion time on the supine and upright seat egress and walk ($r^2 = 0.42–0.47$), followed by ladder climb ($r^2 = 0.41$), recovery from fall ($r^2 = 0.40$)[33]. Additionally, relationships between isokinetic strength and functional task performance were all weaker than those between leg press strength and performance on functional tasks. Taylor et al.[36] found that combined with knee extension total work and knee flexion total work, the lower-body strength measures accounted for 61.5% of the variability (though not significant, $p = 0.15$) in total time to complete the task battery (individual contributions of knee strength parameters were not reported). When performance on additional occupational strength tests (i.e., hand drilling and wrenching strength) was added to the model, 99% of the variability in time to completion could be explained.

Muscular Power: Muscular power is a function of the ability to produce force at high velocities of movement and has consistently demonstrated strong correlations with the performance of tasks of daily living[48,49]. Leg press peak power (normalized to body weight) accounted for the most variance in models for ladder performance ($r^2 = 0.59$), whereas upright seat egress, object carry, and rise from fall were between $r^2 = 0.37–0.49$[33,34]. Bench press peak power (normalized to body weight) accounted for the most variance in models for hatch opening performance ($r^2 = 0.62$)[33]. Results from bed rest and spaceflight data indicated that lower body maximum power was significantly negatively related to performance on seat egress and walk (Somers $D = -0.319$), recovery from fall ($D = -0.361$), and object translation ($D = -0.414$), while upper body maximum power was not significantly correlated with any of the functional fitness tasks[38]. No associations between countermovement jump and function performance were reported[34]. The Wingate test is a common assessment of anaerobic capacity and peak power (presented in Ryder et al.[34] as peak power achieved for 5 s of the 30-s test); however, no studies included in this review reported correlations between Wingate peak power and performance on any functional outcome.

Muscular endurance: Muscular endurance is the ability to perform repeated high-force and velocity movements without fatiguing. Leg press muscle endurance accounted for the most variance in models supine seat egress performance ($r^2 = 0.51$) and ladder climb ($r^2 = 0.56$)[34], and was negatively associated with recovery from fall (Somers $D = -0.462$) and object translation ($D = -0.325$)[38]. Two-minute body weight squats were significantly negatively correlated with 10 km walkback performance ($r = -0.23$), physical abilities test ($r = -0.41$)[28], and material transport field test ($r = -0.36$)[29]. Upper body muscle endurance, as measured by the pushup test, was negatively related to performance on the material transport field test ($r = -0.42$) and device operations field test ($r = -0.59$)[29]. Upper body muscle endurance measured by bench press total work was strongly predictive of hatch opening ($r^2 = 0.73$), but was less predictive of performance on upright seat egress, object carry, and rise from fall ($r^2 = 0.04–0.21$)[33]. Handgrip endurance was only assessed in one study[29], but was not significantly related to performance on the material transport field test or device operations field test.

Thresholds for success/completion time

Strength-Isometric: Thresholds for unilateral isometric knee extension strength in the Ryder et al.[33] study were relatively consistent across functional tasks requiring lower-body involvement (i.e., upright seat egress, rise from fall, supine seat egress, ladder climb, and object carry), ranging from 4.5–5.9 N/kg. Similarly, Ryder et al.[34] reported thresholds for capsule egress, ambulation and supply transfer, crewmember rescue, and hill climb and descent tasks ranging from 5.34 to 5.54 N/kg. Thresholds for leg press isometric strength in Ryder et al.[33] were highest for upright seat egress (17.8 N/kg), followed by similar thresholds for other tasks assessed (13.1–13.9 N/kg). Thresholds for leg press isometric strength in Ryder et al.[34] were similar between all functional tests examined, ranging from 17.3 to 17.65 N/kg. The bench press threshold was ~4× greater for hatch opening (13.0 N/kg) than for object carry (3.4 N/kg) and construction activity board (3.1 N/kg)[33]. Additionally, thresholds for bench press isometric strength in Ryder et al.[34] were similar between all functional tests examined, ranging from 6.39–6.67 N/kg. Only one investigation assessed full-body isometric strength via IMTP[34]. Similar thresholds for IMTP were observed between all functional tests examined, ranging from 18.12 to 18.53 N/kg.

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Table 5 | Muscular fitness metrics

Fitness category	Fitness sub-category	Fitness metric	Thresholds	Associations/relationship to performance outcomes
Strength	Isometric strength	Mid-thigh pull (N/kg)	Capsule egress: 18.12; ambulation and supply transfer: 18.36; crewmember rescue: 18.20 N/kg; hill climb and descent: 18.53 (Ryder et al.[34])	Not reported
		Leg press (N/kg)	Upright seat egress: 17.8; rise from fall: 13.9; supine seat egress: 13.7; ladder climb: 13.3; object carry: 13.1 (Ryder et al.[33])	Supine seat egress: $r^2 = 0.68$; upright seat egress: $r^2 = 0.60$; object carry: $r^2 = 0.52$; ladder treadmill: $r^2 = 0.60$; recovery from fall: $r^2 = 0.54$ (Ryder et al.[33])
			Capsule egress: 17.36; ambulation and supply transfer: 17.51; crewmember rescue: 17.30; hill climb and descent: 17.65 (Ryder et al.[34])	Pre to post changes: seat egress and walk: $D = -0.305$; recovery from fall: $D = -0.363$; object translocation: $D = -0.285$; ladder climb: $D = -0.189$; hatch opening: $D = 0.175$ (Mulavara et al.[38])
		Bench press (N/kg)	Hatch opening: 13.0; object carry: 3.4; construction activity board: 3.1 (Ryder et al.[33])	Hatch opening: $r^2 = 0.67$; object carry: $r^2 = 0.34$; construction activity board: $r^2 = 0.07$ (Ryder et al.[33])
			Capsule egress: 6.40; ambulation and supply transfer: 6.57; crewmember rescue: 6.39; hill climb and descent: 6.67 (Ryder et al.[34]).	No significant associations with seat egress and walk, recovery from fall, object translocation, jump down, ladder climb, activity board, hatch opening (Mulavara 2018)
		Knee extension (N/kg)	Upright seat egress: 5.9; rise from fall: 5.4; supine seat egress: 5.7; ladder climb: 5.2; object carry: 4.5 (Ryder 2013)	Ladder treadmill: $r^2 = 0.52$; supine seat egress: $r^2 = 0.47$; upright seat egress: $r^2 = 0.42$; rise from fall: $r^2 = 0.39$; object carry: $r^2 = 0.37$; not associated with construction activity board or hatch opening (Ryder 2013)
			Capsule egress: 5.44; ambulation and supply transfer: 5.43; crewmember rescue: 5.34; hill climb and descent: 5.54 (Ryder 2019)	
		Handgrip	Not reported	Material transport: $r = -0.49$; device operations: $r = -0.47$ (Ade 2015)
	Isokinetic strength	Concentric knee extension (Nm/kg)	Upright seat egress: 1.9; supine seat egress: 1.8; rise from fall: 1.7; object carry: 1.7; ladder climb: 1.0 (Ryder 2013)	Supine seat egress: $r^2 = 0.47$; upright seat egress: $r^2 = 0.42$; ladder climb: $r^2 = 0.41$; recovery from fall: $r^2 = 0.40$; object carry: $r^2 = 0.27$ (Ryder 2013)
			Capsule egress: 1.83; ambulation and supply transfer: 1.83; crewmember rescue: 1.83; hill climb and descent: 1.86 (Ryder 2019)	When combined with knee extension total work and knee flexion total work, lower-body strength measures accounted for 61.5% of the variability ($p = 0.15$) in total time to complete the task battery (Taylor 2018)
		Concentric knee flexion (Nm/kg)	Capsule egress: 0.93; ambulation and supply transfer: 0.93; crewmember rescue: 0.93; hill climb and descent: 0.94 (Ryder 2019)	Not reported
		Eccentric knee extension (Nm/kg)	Capsule egress: 2.38; ambulation and supply transfer: 2.41; crewmember rescue: 2.40; hill climb and descent: 2.44 (Ryder 2019)	Not reported
		Eccentric knee flexion (Nm/kg)	Capsule egress: 1.40; ambulation and supply transfer: 1.41; crewmember rescue: 1.40; hill climb and descent: 1.43 (Ryder 2019)	Not reported
		Concentric ankle plantarflexion (Nm/kg)	Capsule egress: 1.21; ambulation and supply transfer: 1.21; crewmember rescue: 1.21; hill climb and descent: 1.22 (Ryder 2019)	Not reported
		Eccentric ankle plantarflexion (Nm/kg)	Capsule egress: 1.76; ambulation and supply transfer: 1.77; crewmember rescue: 1.76; hill climb and descent: 1.78 (Ryder 2019)	Not reported
		Concentric ankle dorsiflexion (Nm/kg)	Capsule egress: 0.33; ambulation and supply transfer: 0.33; crewmember rescue: 0.33; hill climb and descent: 0.33 (Ryder 2019)	Not reported
		Eccentric ankle dorsiflexion (Nm/kg)	Capsule egress: 0.58; ambulation and supply transfer: 0.57; crewmember rescue: 0.58; hill climb and descent: 0.58 (Ryder 2019)	Not reported
		Trunk extension (Nm/kg)	Capsule egress: 3.79; ambulation and supply transfer: 3.80; crewmember rescue: 3.78; hill climb and descent: 3.84 (Ryder 2019)	Not reported
		Trunk flexion (Nm/kg)	Capsule egress: 1.88; ambulation and supply transfer: 1.88; crewmember rescue: 1.90; hill climb and descent: 1.89 (Ryder 2019)	Not reported
Power	Lower body power	Isotonic leg press (W/kg)	Upright seat egress: 17.6; supine seat egress: 14.6; object carry: 6.5; ladder treadmill: 8.2; rise from fall: 8.2 (Ryder 2013) Capsule egress: 14.82; ambulation & supply transfer: 14.91; crewmember rescue: 14.80; hill climb & descent: 15.10 (Ryder 2019)	Upright seat egress: $r^2 = 0.40$; supine seat egress: $r^2 = 0.49$; object carry: $r^2 = 0.37$; ladder treadmill: $r^2 = 0.59$; rise from fall: $r^2 = 0.45$ (Ryder 2013)

Table 5 (continued) | Muscular fitness metrics

Fitness category	Fitness sub-category	Fitness metric	Thresholds	Associations/relationship to performance outcomes
		Countermovement Jump (W/kg)	Capsule egress: 31.16; ambulation & supply transfer: 31.46; crewmember rescue: 31.12; hill climb & descent: 31.86 (Ryder 2019)	Not reported
		Cycle Wingate (W/kg)	Capsule egress: 5.74; ambulation and supply transfer: 5.78; crewmember rescue: 5.68; hill climb and descent: 5.83 (Ryder 2019)	Not reported
	Upper body power	Isotonic bench press (W/kg)	Object carry: 8.2; hatch opening: 10.8; construction activity board: 2.3 (Ryder 2013) Capsule egress: 6.22; ambulation & supply transfer: 6.41; crewmember rescue: 6.26; hill climb & descent: 6.49 (Ryder 2019)	Object carry: $r^2 = 0.18$; hatch opening: $r^2 = 0.62$; construction activity board: $r^2 = 0.04$ (Ryder 2013)
Endurance	Lower body endurance	2-min squats	Not reported	10 km walkback: $r = -0.23$, physical abilities test: $r = -0.41$ (Ade 2014) Material transport field test: $r = -0.36$; device operations: $r = -0.25$ (Ade 2015)
		Isotonic leg press (J/kg)	Upright seat egress: 78.8 supine seat egress: 72.4; object carry: 71.4; ladder treadmill: 34.4; rise from fall: 71.4 (Ryder 2013) Capsule egress: 74.73; ambulation & supply transfer: 75.55; crewmember rescue: 73.81; hill climb & descent: 76.76 (Ryder 2019)	Supine seat egress: $r^2 = 0.51$; ladder treadmill $r^2 = 0.56$ (Ryder 2013) Recovery from fall: $D = -0.462$; object translation: $D = -0.325$ (Mulavara 2018)
		Isokinetic knee extension/flexion (J/kg)	Capsule Egress: 22.9 (extension)/12.78 (flexion); ambulation & supply transfer: 22.98/12.85; crewmember rescue: 22.84/12.75; hill climb & descent: 23.34/12.98 (Ryder 2019)	Not reported
	Upper body endurance	Handgrip endurance	Not reported	Material transport: $r = -0.13$; device operations: $r = -0.38$ (Ade 2015)
		Isotonic bench press (J/kg)	Object carry: 18.3; hatch opening: 46.0; construction activity board: 9.9 (Ryder 2013) Capsule egress: 25.26; ambulation & supply transfer: 26.11; crewmember rescue: 25.08; hill climb & descent: 26.50 (Ryder 2019)	Hatch opening: $r^2 = 0.73$; object carry: $r^2 = 0.21$; construction activity board: $r^2 = 0.05$ (Ryder 2013)
		2-min pushups	Not reported	Material transport: $r = -0.42$; device operations: $r = -0.59$ (Ade 2015)

Thresholds were identified as specific values determined through regression or receiver operator curves as being related to task performance. Relationships were reported as r, r^2, or Somers' D values.

Strength-Isokinetic: Ryder et al.[34] reported similar isokinetic strength thresholds across muscle groups for each functional task assessed, including those reported for concentric knee extension (1.83–1.86 Nm/kg), eccentric knee extension (2.38–2.44 Nm/kg), concentric knee flexion (0.93–0.94 Nm/kg), eccentric knee flexion (1.40–1.43 Nm/kg), concentric ankle plantarflexion (1.21–1.22 Nm/kg), eccentric ankle plantarflexion (1.76–1.78 Nm/kg), concentric ankle dorsiflexion (0.33 Nm/kg), eccentric ankle dorsiflexion (0.57–0.58 Nm/kg), trunk extension (3.78–3.84 Nm/kg), and trunk flexion (1.88–1.90 Nm/kg). Thresholds for concentric knee extension in Ryder et al.[33] showed more dispersion and were highest for upright and supine seat egress (1.8–1.9 Nm/kg), followed by rise from fall and object carry (1.7 Nm/kg), and then ladder climb (1.0 Nm/kg).

Muscular Power: Performance thresholds for leg press peak power reported in Ryder et al.[34] were comparable across functional tasks (i.e., capsule egress, ambulation, and supply transfer, crewmember rescue, and hill climb and descent), ranging from 14.8 to 15.1 W/kg. Countermovement jump power thresholds were larger than leg press power values, ranging from 31.12 to 31.86 W/kg for the same tasks. Performance thresholds for bench press peak power reported in Ryder et al.[34] were comparable for the same tasks, ranging from 6.22 to 6.49 W/kg. Only one study investigated Wingate peak power and values for capsule egress, ambulation and supply transfer, rescue drag, and hill climb and descent were closely related (range 5.58–5.83 W/kg)[34].

Muscular Endurance: Performance thresholds for muscular endurance were comparable across functional tasks (i.e., capsule egress, ambulation and supply transfer, crewmember rescue, and hill climb and descent): leg press endurance (73.81–76.76 J/kg), isokinetic knee extension (22.84–23.34 J/kg), isokinetic knee flexion (12.75–12.98 J/kg), bench press (25.08–26.50 J/kg)[34].

Simulation reported efforts

To determine the fitness level required for future exploration mission surface operations, physiological workload documented in EVA simulation studies and standalone tasks may provide additional insight.

Offload simulations/standalone tasks: In offload environments (NASA's Partial Gravity Simulator POGO, simulating 1/6 g [Lunar] and 3/8 g [Martian]), standalone tasks, particularly those performed in Martian offload, were performed at high metabolic effort. When suited in 1/6 g, treadmill ambulation efforts ranged from 16.7–29.3 mL/kg/min, while 3/8 g suited condition yielded efforts ranging from 27.6–35.7 mL/kg/min[39]. For a 10 km walkback task in the Mark III suit at 1/6 g offload, average VO_2 values of 2.0 ± 0.3 L/min, which equated to 50.8 ± 6.1% VO_{2pk} (range 45–61% VO_{2pk})[39]. Average rate of perceived exertion was 11.8 ± 1.6 out of 20, which corresponded to "light" to "somewhat hard" categorization[39]; however, it is unknown how intensity (both quantified at percentage of aerobic capacity or perceived exertion) would compare when performed at a Martian offload.

1 g simulations/standalone tasks: Several studies performed in 1 g (i.e., without gravitational offloading) reported metrics of fitness and their relationship to simulation efforts and standalone EVA task performance. Similar to offloaded investigations, simulations performed at Martian intensities required greater fitness than Lunar efforts. Sutterfield et al.[35] demonstrated that treadmill traverse performed in 1 g at Lunar intensities averaged 24.0 ± 1.8 mL/kg/min, equating to approximately ~67% of VO_{2pk}, while the Martian traverse averaged 28.7 ± 1.8 mL/kg/min, equating to approximately ~80% VO_{2pk}. Interestingly, failure rates for Martian traverse were more than 2× that of Lunar failure rates (38.5% vs. 13%)[35]. Additionally, a Lunar intensity 2 km hill climb was performed at 20 ± 2.3 mL/kg/min (~56% VO_{2pk}) and a Martian hill climb at 26.0 ± 0.7 mL/kg/min (~73% VO_{2pk}), where again a greater failure rate was seen with the Martian intensity task (25% vs. 13%)[35].

In two studies that examined self-paced 2 h 1 g EVA simulations in a Mars analog suit designed to mimic geological surveying, efforts yielded average intensities of 35.9 ± 0.9 mL/kg/min[37] and 81–85% of heart rate max[32,37]. VO_2 was predictive of distance covered during EVA ($r^2 = 0.34$), with greater oxygen demand per time period for fitter subjects ($r^2 = 0.86$)[37].

Critical appraisal for study and reporting quality

Using the AXIS tool, each study (Fig. 3) included clear study objectives, target populations, and outcomes that seemed appropriate to meet the aims.

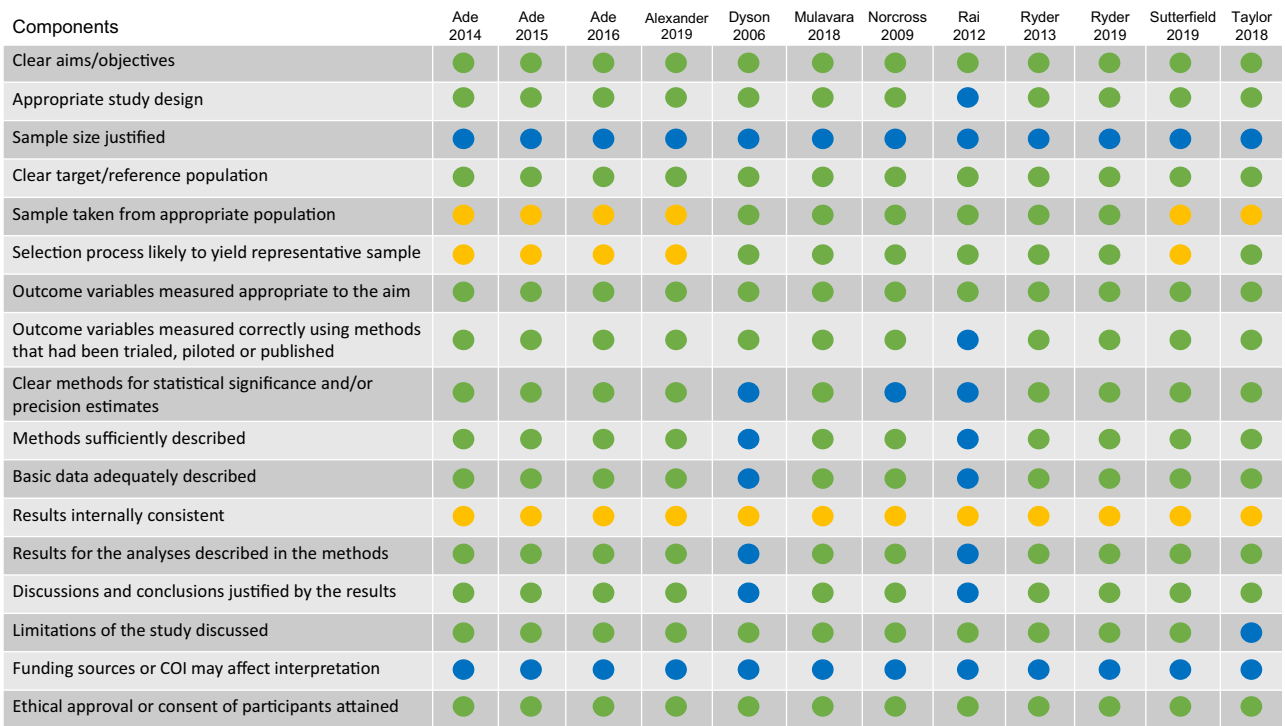


Fig. 3 | AXIS critical appraisal (Downes et al.[13]). Critical appraisal outcomes for included studies. Green indicates "yes", blue indicates "no", and yellow indicates "do not know" for each AXIS component.

There were no indications of funding sources or conflicts of interest that would impact interpretation. Of these studies, 10 of 12 studies yielded assessments that indicated that study design, methods, and results were reported in a manner reflective of a quality investigation. While effect size and power estimates were not reported for any studies, information provided indicated that objectives, study design, and methodology were sufficiently described and appropriate to draw conclusions from the results reported. Two of the 12 studies did not report methods or results in a way that yielded confidence in findings. Certain methods included (i.e., VO_{2max} assessed via a hill climb protocol[32]) are not considered appropriate and therefore impact the interpretation of intensity of a Mars simulated EVA.

Discussion

This systematic review aimed to consolidate findings to identify which components of fitness were most related to EVA task and simulated EVA performance. From the 12 studies included in the review, 13 cardiorespiratory and 25 muscular fitness metrics were identified. Of the cardiorespiratory metrics, 8 were associated with performance. Treadmill VO_{2max} and treadmill CS had the strongest association with EVA task and simulated EVA performance, where greater fitness was related to greater performance (i.e., quicker task completion times). Of the muscular metrics, 12 were associated with performance, with isometric leg press and bench press yielding the strongest predictive values of muscular strength measures and isotonic bench press yielding the strongest predictive value of endurance measures. Findings presented herein focus on 1 g participants and provide evidence that is predominantly associative in nature and appears to be driven by the task and context tested. While limitations of extending 1 g findings to future exploration settings exist, the identification of these key fitness metrics may provide useful information for refinement of NASA-STD-3001 and the development of future exploration exercise capabilities and monitoring systems.

Of the key physiologic factors that influence endurance exercise performance, aerobic capacity (VO_{2pk} or VO_{2max})[40] was the most common aerobic fitness metric, examined in 7/12 studies with EVA task circuits or standalone tasks. VO_{2pk} assessed via treadmill demonstrated the strongest relation to task performance compared to other methods (i.e., arm crank)[28,29], particularly for ambulatory tasks like 10 km walkback. Additionally, several studies determined VO_{2pk} thresholds for task completion when performed at different intensities (e.g., Lunar [low: 25.7–26.1 mL/kg/min] vs Martian [moderate: 35–40 mL/kg/min] intensity)[28,29,35]; however, more research is needed to confirm that these intensities are representative of exploration EVA metabolic rates. Submaximal indicators of effort, such as VT (or GET), also demonstrated associations with EVA task performance. Because crewmembers may be limited by suit mobility, consumables conservation, and operations planning to prioritize safety and mission success, it is unlikely that prolonged maximal intensity efforts will be seen, unless in contingency scenarios. Therefore, submaximal workload indicators may be a candidate fitness metric of interest for further investigation. As both aerobic capacity and VT/GET require metabolic assessment for accuracy, research and testing involving portable gas exchange systems suitable for in-flight use in different gravitational environments is warranted.

Of the fitness assessments, CS (via treadmill) and CP (via arm crank) demonstrated the strongest relation to EVA task performance[28,29], while other investigators identified thresholds for task success using rowing CP[35]. Because exploration EVAs are planned to be of extended duration (i.e., 4–8 h in length), these indicators of exercise tolerance and endurance[43,45,46] may be of great interest; however, the considerations for applicability for fitness monitoring are dependent on balancing crew time, availability of in-flight exploration resources, and vehicle constraints. In particular, CP assessed via rowing[35] presents a viable candidate for in-flight fitness monitoring that maximizes efficiency and limits consumables. Rowing CP can be completed in less than 10 min (warmup, 3-min "all-out" effort, and cool down)[35] and could be performed without metabolic gas analysis, if metabolic monitoring was not available. Additionally, CP could be adapted for use on exploration exercise countermeasures or assessed via a cycle ergometer, increasing the operational applicability for fitness monitoring pre-, in-, and post-mission. Assessing the feasibility of performing CP within an exploration vehicle, including the determination of loads and dynamics to the vehicle, would be necessary prior to implementation. Conversely, while treadmill CS had the greatest association reported among all cardiovascular fitness metrics, the current utility of the test in an astronaut population may be limited. Because deconditioned crewmembers may be most vulnerable to injury early post-flight, treadmill running may not be advisable due to sensorimotor disruptions. Additionally, the assessment of CS described in Ade et al.[28], required 4 separate bouts of running at speeds ranging from 90 to 120% VO_{2pk} (ranging 2–15 min) on 4 separate occasions, which would not be feasible with respect to crew time and scheduling. Importantly, due to size, power, and mass constraints of vehicles, a treadmill is not currently planned as an exercise countermeasure on future exploration missions.

Data from the current manuscript indicate that tests involving ambulation (i.e., treadmill VO_{2pk}, VT, CS) are related to simulated task performance in 1 g subjects. Additionally, in crewmembers, treadmill-based exercise confers unique benefits to the neurovestibular and sensorimotor systems[50,51] and may positively impact EVA performance. Together, these findings highlight the utility of ambulatory-based assessments and exercise, supporting the need for treadmills to be included in the countermeasure systems of long-duration missions. Innovative treadmill designs that fit within the spacecraft mass, volume, and power constraints are urgently needed to ensure the countermeasure modalities needed to maintain crew health and performance are available during extended-exploration missions. Research is needed, and is currently underway, to determine the magnitude of impact that the lack of treadmill exercise has on crewmember health and performance, and may yield further data to justify a treadmill for exercise and monitoring purposes in future exploration missions.

Regarding lower body strength assessments most related to EVA task performance, isometric strength, particularly for leg press, demonstrated the strongest relationship to task performance. This relationship was strongest for tasks requiring lower body effort, such as seat egress and ladder climb[33]. Leg press isometric strength thresholds for task success range from 13.1 to 17.8 N/kg and are dependent on task[33,34]. For upper body strength assessments, isotonic bench press for endurance had the strongest relationship to hatch opening, but weaker relationships were seen with other upper body tasks, such as object carry or activity board[33]. Thresholds for isotonic bench press ranged from 25.08 to 26.50 J/kg[34]. Isometric testing is reliable and safe[47] and has also been a part of the NASA research testing battery[52] to assess changes in astronaut strength pre- to post-flight. Recently, IMTP has been included as a measure of in-flight strength being tested on current exercise hardware in several ongoing flight studies. As such, isometric assessments represent a favorable form of muscular testing that could be utilized for exploration mission fitness characterization.

Interestingly, while 1-RM assessments are used in strength standards for NASA crewmembers, no investigations included in this review examined 1-RM measures in relation to task performance. While 1-RM testing is considered the non-laboratory "gold standard" for dynamic strength assessments[53] and generally regarded as safe across terrestrial populations[54], spaceflight deconditioning can lead to spinal changes and atrophy of intrinsic spinal and lumbar paraspinal muscles[55], potentially increasing the risk of injury in returning crew[56] and possibly preventing the performance of a true maximum effort. Because these 1-RM standards had not previously been tested for their utility as performance indicators of EVA and were not included in any of the studies included in this literature review, understanding the relevance of these assessments in this population, who may be at increased risk of injury, is essential. While 1-RM measures were not assessed in the studies included in this literature, numerous other muscular metrics (e.g., isometric strength for lower body [leg press], upper body [bench press], and whole body [mid-thigh pull]), demonstrated strong relationships or predictive capabilities for simulated EVA task performance in terrestrial subjects, warranting further exploration into the potential utility for NASA standards with microgravity exposed crewmembers.

It is apparent that the relationship between fitness indicators and task performance is dependent on the design of the task/test and is an important consideration when interpreting the findings presented in this review. Designing a test or circuit that incorporates "failure" is one approach taken by several of these research investigations. Pushing subjects to failure allows for a better comparison of "maximal" fitness tests to "maximal" task performance; however, sequential back-to-back maximal tasks as part of a circuit may not be as representative of what might occur during EVAs due to suit metabolic, environmental, and consumable limits, which is a limitation of the current research presented herein. Using Apollo missions for reference, the average metabolic expenditure for Apollo 11-17 Lunar EVAs[57] was 906 BTU/h (~11 mL/kg/min), with higher metabolic rates reported for tasks like a geological station activity (1393 BTU/h; ~16 mL/kg/min), ALSEP deployment (1297 BTU/h; ~15.5 mL/kg/min), and overhead task (1167 BTU/h; ~15 mL/kg/min), representing relatively low workloads. However, there were a few instances of heavier workloads. During EVA 1 of Apollo 16, the Lunar module pilot (LMP) carried, dropped, and re-assembled 2 large packages (21 pounds lunar weight each) during a total duration of 4.62 min[58]. The metabolic rate increased to 2300 BTU/h (~24 mL/kg/min) when the LMP carried the re-assembled packages 180 feet in 60 s, one of the highest recorded during Apollo missions, but quickly recovered to 1282 BTU/h (~16 mL/kg/min) within 3 min[58]. This example demonstrates that short, intense efforts with a rest between tasks may be more representative of EVA activity, as opposed to sustained intense effort to failure, for durations ranging from 20 to 36 min, as was employed by several of the task circuit studies[28,29]. Additionally, as EVAs represent one of the most perilous undertakings of spaceflight exploration[7], performance on EVAs often includes methodical and controlled movements to reduce the risk of injury and decompression sickness, and workload intensities are limited by suit consumables and environmental control conditions; therefore, short-duration, high-intensity task circuits could far exceed the metabolic demand experienced during surface operations. Ultimately, tasks selection for the determination of minimum fitness required for surface EVAs is critical; understanding how tasks selected relate to expected EVA performance requires higher fidelity investigations in analog environments for a more realistic representation of end-to-end EVA.

Additional limitations to this study exist. Many studies presented data for correlations between fitness and task performance, but associations do not indicate whether improvements in fitness level will definitively lead to improvements in EVA performance. Heterogeneity existed with respect to EVA tasks tested and duration of assessment, which resulted in a wide range of values identified as thresholds for certain tasks and the inability to perform a meta-analysis of correlations. Beyond the standalone traverse task performed in offloaded conditions in a single study[39], the fitness required to perform other operational tasks in offloaded gravitational conditions, over an extended period of time, is unknown. Similarly, most task circuit studies included were performed in shirt-sleeve conditions on level surfaces, which do not account for potential biomechanical constraints and limitations imparted by the pressurized spacesuit[59] or added metabolic cost of uneven terrain[14], which may alter and influence movement strategies needed to successfully complete tasks. Because few laboratories exist around the world to allow for these higher-fidelity investigations (i.e., appropriate offloading or suited testing), the research presented herein represents the best indication of how fitness may influence EVA task performance, given the obvious limitations. While this review focused on specific domains of fitness in relation to task performance in a bivariate manner, fitness is not the sole predictor of EVA performance, with other physiological systems impacted by microgravity exposure, change in gravitational fields, or influence of high-stress environment playing contributing roles.

In summary, results from this systematic review highlight that aerobic capacity, lower body strength (leg press), and upper body endurance and strength (bench press) yield the strongest relationships to EVA task performance studied in terrestrial populations, mostly in low-fidelity environments (1 g weighted suit, 1 g unsuited, etc). These findings may help inform revisions to the NASA-STD-3001 and provide information for consideration as exercise countermeasure and monitoring systems are developed for exploration missions. Ultimately, to best address how fitness will impact surface EVAs on the Moon and Mars, higher fidelity investigations, such as those including Lunar or Martian gravity offloading, are warranted.

Other information

This review was not registered.

Data availability

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Code availability

Code sharing is not applicable to this article.

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Author contributions

N.S. led the literature search, critical appraisal, and manuscript writing. A.V. was a major contributor to the literature search, critical appraisal, and manuscript writing and review. J.A. was a major contributor to literature search, critical appraisal and manuscript review. D.F. contributed to publication search via alternative methods (NTRS and hand search), data extraction, and manuscript review. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Nicole C. Strock.

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